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The Radiologic Anatomy of the Lumbar and Lumbosacral Pedicles

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Study Design. An anatomic and radiologic study of lumbar and lumbosacral pedicle anatomy.

Objectives. To define the radiologic anatomy of the lumbar and first sacral pedicle in the coaxial projection.

Summary of Background Data. Fluoroscopic assistance for pedicle screw placement requires radiologic landmarks. The radiologic landmarks have previously been assumed. Detailed study of the correlation between anatomy and radiology is required.

Methods. Lumbar vertebrae and sacra were marked with radiopaque material to demonstrate the pedicle cortical borders. The vertebrae were then imaged in the coaxial projection to determine the correlation between the pedicle cortex and the radiologic image. Pedicle dimensions were recorded.

Results. Pedicle dimensions were consistent with known measurements, yet the long axis of the L4 and L5 pedicle ellipse was oblique to the vertical. Consequently, the minor diameter of the pedicle ellipse was considerably less than the measured pedicle width at L5. The radiologic pedicle image was consistently within the true pedicle cortex, by up to 3 mm, and probably represents the inner cortical border of the pedicle. The S1 pedicle has reliable anatomic landmarks, yet only the medial and superior borders were visualized.

Conclusions. The radiologic pedicle image in the lumbar and lumbosacral spine is a reliable guide to the true bony cortex of the pedicle. At S1 the pedicle image is less well correlated with the cortical borders of the pedicle, yet other reliable anatomic landmarks exist. [Key words: lumbar pedicle anatomy, pedicle radiology, pedicle screws]
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The mechanical advantages provided by pedicle screw fixation systems have resulted in their widespread use in the thoracic, lumbar, and sacral regions.[44] Concern regarding safety, potential complications if screws are misplaced,[6,10,12,14,42,43,45] and loss of mechanical advantage with pedicle wall disruption[15,23,25,30] has focused attention on screw placement techniques. Freehand screw placement techniques are associated with the risk of screw misplacement,[32,40,41] and routine posteroanterior and lateral radiographs are unreliable for assessing screw position.[13,41] Neurophysiologic monitoring has been used to assist safe screw placement.[4,9,16,19,28–30,39] Computer-assisted guidance systems offer promise[5,26,37] when technical difficulties are overcome.[17,18,21] Both these techniques require special equipment with associated expense. Intraoperative fluoroscopy coaxial to the pedicle has been well described[24] and is accurate in cadaveric specimens when pedicles are of adequate size to accept a screw.[38]

The anatomy of the thoracic, lumbar, and sacral pedicles has been well defined in cadaveric and radiologic studies.[2,3,25,33,34,46] Definition has included dimensions of the pedicle and the pedicle axes. The anatomic dimensions have taken into account age and racial variation.[7,20,22] Conversely, there has been no description, of which the authors are aware, in which the definition of pedicles in terms of shape and dimension within the coaxial image (*i.e.*, perpendicular to the pedicle axis) has occurred. It is conceivable that anatomic nuances when viewed from this perspective may alter understanding of pedicle anatomy. It is also unclear what radiologic landmarks should be used to assess safe and appropriate placement of screws within the pedicle.

The purposes of this anatomic study were to define pedicle dimensions and shape when viewed in the coaxial projection in the lumbar and lumbosacral spine and to define the relation between the cortical bone and the radiologic image when using coaxial imaging, to provide guidelines for safe anatomic boundaries when using fluoroscopy-assisted pedicle screw placement.

Methods

Ten dry-bone lumbosacral spine specimens were studied, for a total of 100 lumbar pedicles and 20 S1 pedicles. The isthmus (narrowest circumference) of the pedicles of each lumbar vertebrae was identified, and fine, malleable wire (0.4-mm tinned copper wire) was applied to this region. The wire was drawn tightly on the pedicle isthmus so that direct contact existed between the wire and the cortical bone at all points (Figure 1). Initial radiographic studies demonstrated that the fine wire used to define the outer cortical border of the lumbar pedicles did not adequately demonstrate the border of the S1 pedicle. A wider gauge malleable wire (0.8-mm enameled copper wire) was used to define this cortical bony anatomy (Figure 2). The vertebral specimens were then mounted with malleable adhesive on plate glass to enable variable positioning of the specimen within a radiograph beam. Direct posteroanterior radiographs of each vertebra were taken (Figure 3), and then each vertebra was rotated about its longitudinal axis with multiple radiographic images taken until each pedicle image was viewed in its maximum transverse dimension with the whole of the pedicle superimposed within the vertebral body. This is the coaxial image[24] (Figure 4). The specimens were rotated to achieve coaxial images of both pedicles of each specimen.

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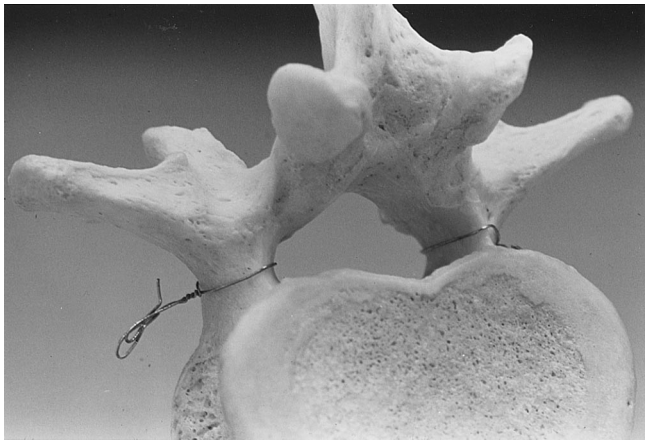


Figure 1. Midlumbar vertebral specimen demonstrating the application of fine, malleable wire to the outer bone surface of the pedicle. The wire is drawn tightly against the outer border of the pedicle.

Pedicle Dimensions. For each pedicle coaxial view the pedicle dimensions were recorded by direct measurements from the radiographs. The pedicle height is the distance between the upper and lower extent of the pedicle cortex, defined by the inner margin of the circumference of pedicle wire, when a pedicle is viewed perpendicular to the vertebral body longitudinal axis on the coaxial view (and therefore excludes effects of the lumbar lordosis). The pedicle width is the dimension from the most medial to lateral extent of the pedicle (as defined by the circumferential pedicle wire) on the coaxial view. The pedicle elliptical axis is the longitudinal axis of the elliptical (approximated ellipse) coaxial pedicle image. The pedicle isthmus diameter is the measurement of the short axis of the pedicle ellipse at the pedicle isthmus. This represents the narrowest diameter within which a pedicle screw can be fitted. All radiologic measurements were corrected in relation to a known standard dimension to eliminate radiologic magnification.

Radiologic Cortical Anatomic Description. Descriptive analysis of the radiologic pedicle image and its relation to the true cortical bony anatomy (defined by the circumferential radiopaque marking wire) of the pedicle was made for the coaxial view. This included analysis of all the lumbar pedicles and the S1 pedicles.

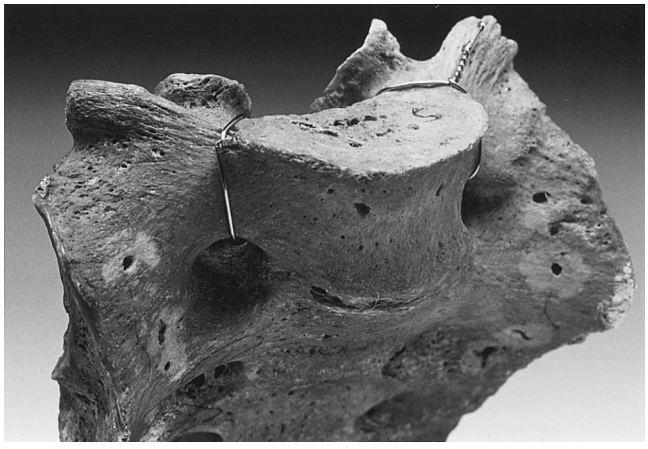


Figure 2. Sacrum specimen demonstrating the application of malleable wire to the outer border of the S1 pedicle. The wire has been drawn tightly against the outer border of the pedicle. Wider gauge of wire was required at the sacrum to allow visualization in the radiologic studies.

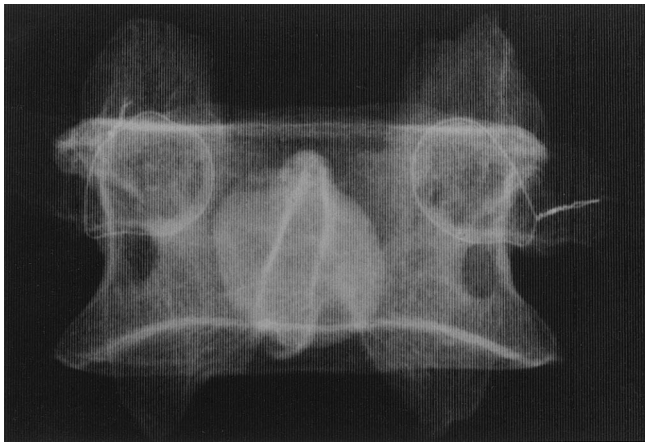


Figure 3. Posteroanterior radiograph of an L4 vertebra with the wire applied to the outer borders of the pedicles. Note that the opaque image overlying the spinous process in this and subsequent images is the adhesive for positioning. Right and left sides are as viewed by the reader.

■ Results

The technique of using fine, malleable wire tightly applied to the outer cortical margin of the pedicle isthmus allowed the outer border of the pedicle to be defined radiologically. Close adherence of the wire to the bone was possible in all situations, except where defects in the cortex associated with vascular foramens were present. These defects in the pedicle cortex were most evident along the superior border of the L5 pedicle.

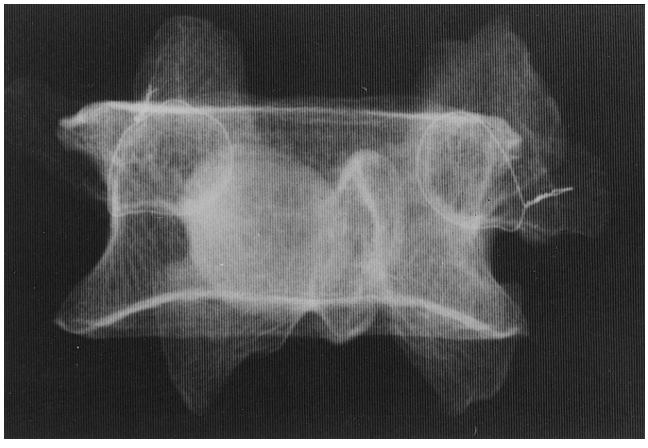


Figure 4. Coaxial radiographic image of the same L4 vertebra with the vertebra rotated around the longitudinal axis to demonstrate the left pedicle (reader's left). The pedicle image is seen to lie within the image of the vertebral body, below the superior endplate of the vertebra and medial to the lateral wall of the vertebral body. The pedicle is seen in its widest projection, and the rotation turns the spinous process away from the imaged pedicle. The opposite pedicle image is narrowed.

Table 1. Coaxial Pedicle View Anatomical Data

		Left		Right	
Level		Mean	STD Dev	Mean	STD Dev
L1	Height	18.6 mm	3.2	19.9 mm	3.9
	Width	10.3 mm	2.5	9.5 mm	2.5
	Axis	−2.4°	5.3	1.7°	3.1
	Isthmus	9.9 mm	2.5	8.7 mm	2.4
L2	Height	18.6 mm	3.0	18.7 mm	3.2
	Width	10.3 mm	3.0	10.0 mm	2.9
	Axis	0.8°	5.6	4.0°	6.1
	Isthmus	10.0 mm	3.0	9.6 mm	2.8
L3	Height	18.6 mm	13.0	18.7 mm	2.5
	Width	13.0 mm	2.6	12.7 mm	2.9
	Axis	3.5°	4.6	7.5°	6.6
	Isthmus	12.3 mm	2.1	12.0 mm	2.9
L4	Height	17.8 mm	3.0	17.4 mm	2.5
	Width	16.3 mm	3.7	16.5 mm	3.8
	Axis	31.0°	17.1	33.5°	20.8
	Isthmus	13.4 mm	2.8	12.2 mm	2.1
L5	Height	17.6 mm	2.9	19.5 mm	3.3
	Width	21.6 mm	3.9	21.5 mm	4.5
	Axis	55.0°	8.8	50.3°	7.4
	Isthmus	13.2 mm	2.4	14.0 mm	2.9

Coaxial Pedicle Dimensions

The data that were obtained for the pedicle anatomy in the coaxial plane are represented in Table 1. These results confirm the previous observations that the pedicle width increases dramatically when progressing from L1 to L5. Although the coaxial view of the pedicle shows an ellipse with a vertical long axis through most of the thoracic and lumbar spine, this study shows that the pedicle elliptical axis becomes oblique at L4 and L5 (left L4 axis 31.0°, L5 axis 55.0°; Figure 5). Although pedicle width is greater in the distal lumbar spine, this increased width is not represented by as great an increase in the pedicle isthmus diameter (the key parameter when assessing whether the pedicle screw will fit within the pedicle walls[1]) because of the obliquity of the pedicle elliptical axis from the vertical.

Radiologic–Cortical Anatomy Correlation

The cortical image of each pedicle from L1 to L4 was fully visualized in all cases to lie within the wire defining the outer border of the pedicle on each specimen. At L5 the superior, medial, and inferior borders were also visualized, yet the lateral border of the L5 pedicle was superimposed on the image of the lateral border of the body of L5. The radiologic image of the pedicle was of maximum density 1–3 mm within the wire image, suggesting that the inner rather than the outer cortex of the pedicle is what was well visualized radiologically (Figure 6). Indeed, in some cases the wire appeared to be providing a halo image to the pedicle. This occurred even though inspection of the specimen confirmed the wire to be in direct contact with the cortical bone of the pedicle. This suggests that the maximum density of the radiologic image of the pedicle is well within the true cortical boundary of the pedicle.

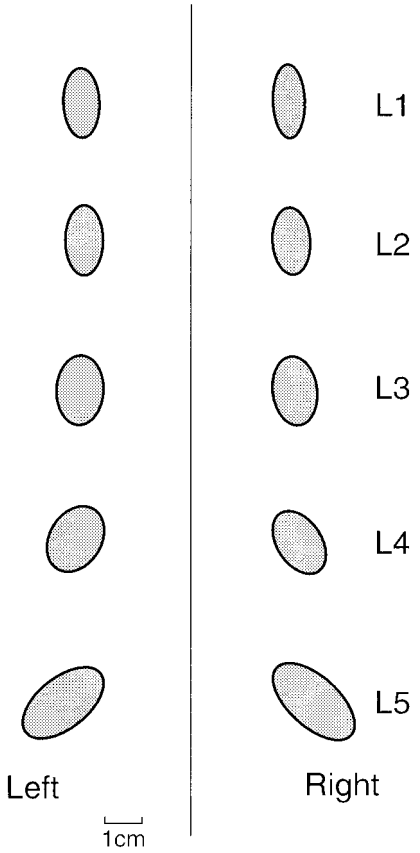


Figure 5. Representation of the pedicle size and shape in the lumbar spine, when the pedicles are viewed in the coaxial plane. Note that the upper lumbar pedicle shape conforms to the known dimensions. At L5 the pedicle has an elliptical shape where the long axis has rotated away from the vertical, so that although it has a large pedicle height and width, the pedicle isthmus diameter (the effective pedicle diameter in which a screw would fit) is much narrower than the medial-to-lateral dimension of the L5 pedicle.

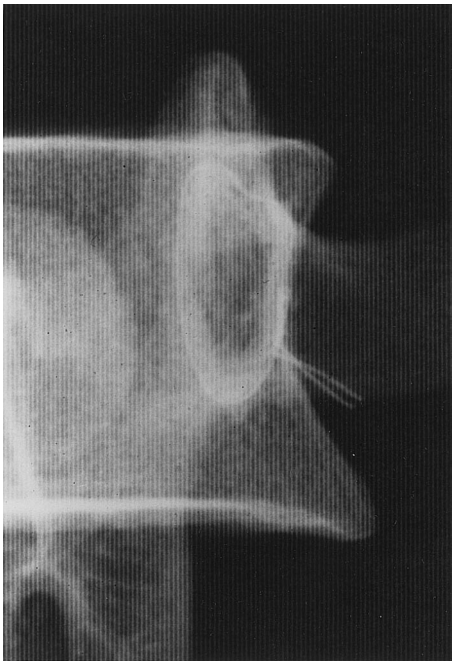


Figure 6. Close-up view of the coaxial image of a midlumbar pedicle. The wire defining the outer border of the pedicle is outside the dense cortical image of the pedicle. The cortical image of the pedicle lies within the true pedicle border, defined by the wire marker.

The base of the transverse process at the junction with the pedicle is most often equivalent in the anteroposterior and coaxial images of the pedicle. As the transverse process develops laterally from the pedicle, its direction in a cephalad and caudad direction is variable. On some occasions the midpoint of the base of the transverse process did not correlate with the center of the pedicle in comparison with its cephalad–caudad position (Figure 7).



Figure 7. Close-up view of the coaxial image of a midlumbar pedicle. In this specimen, the base of the transverse process is at the lower aspect of the pedicle. If the midpoint of the transverse process were used as a guide to the vertical midpoint of the pedicle, the screw would be placed below the center of the pedicle in a suboptimal position.

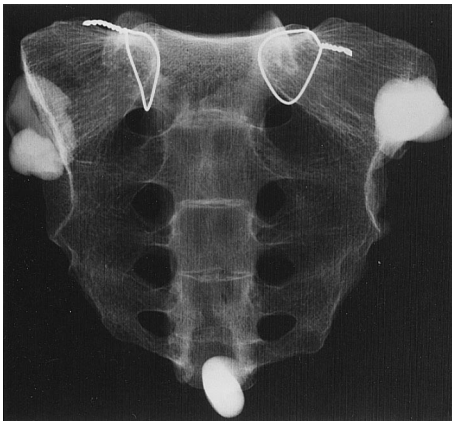


Figure 8. Posteroanterior view of a sacral specimen with mild rotation to "widen" the S1 pedicle on the right.

The pedicle of S1 differs from the lumbar pedicles, because of the confluence of the sacral ala with the lateral aspect of the pedicle. When viewed from above, and in axial projection, the S1 pedicle has a normal lumbar-like medial border, but a lateral aspect that joins with the ala of the sacrum. When the wire marker was applied to the S1 pedicle, the lateral border could be seen only with marked rotation of the sacrum out of the posteroanterior plane. Such rotation would be inappropriate in any clinical situation, and therefore only 10–15° of obliquity was created to "widen" the S1 pedicle image (Figure 8).

With this rotated image of the sacrum, each S1 pedicle was found to demonstrate a triangular shape with superior medial and lateral boundaries. The superior and medial cortices were well seen, yet the lateral cortex was not. The superior cortical image of the S1 pedicle was closely applied to the wire marker. The medial cortical image of the S1 pedicle was placed laterally to the wire (within the pedicle as marked by the wire) in all cases, often by 3–4 mm (Figure 9). This deviation of the radiologic cortical image from the true bone cortical border of the medial pedicle became greater at the caudal extent of the S1 pedicle. Close examination of this radiologic medial border of the S1 pedicle showed it to be a trabecular line that continued sweeping in a lateral and caudad direction out to the sacroiliac joint. No inferior curvature to represent the inferior angle of the S1 pedicle was seen, yet the S1 foramen is clearly evident, and the upper margin of this defines the inferior extent of the S1 pedicle.

The lateral border of the S1 pedicle was not defined using the modest rotation of the sacrum that would be practical in the clinical situation. The intervertebral foramen at S1–S2 was consistently medial to the lateral border of the S1 pedicle. In all cases, this S1–S2 intervertebral foramen represented a valuable landmark

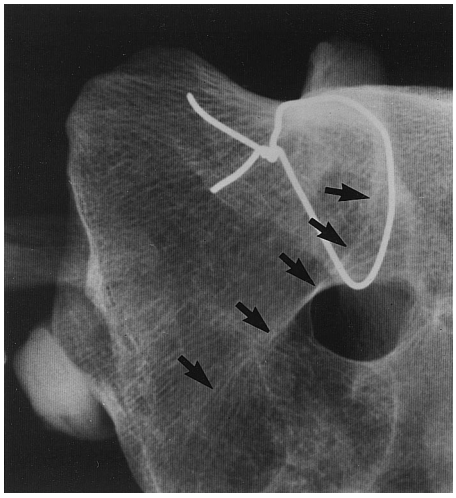


Figure 9. Close-up view of the S1 pedicle. The pedicle boundary is defined by the marking wire. In this view, the superior and medial borders of the triangular S1 pedicle can be seen, but there is no cortical image of the lateral wall of the S1 pedicle. The lateral wall of the pedicle (demonstrated by the wire) lies lateral to the S1–S2 intervertebral foramen. Note that the medial cortical image of the S1 pedicle is a trabecular line (arrowheads) that lies several millimeters within the true medial border of the S1 pedicle (demonstrated by the wire), and this trabecular line continues out to the sacroiliac joint.

from which to define the position of the lateral border of the S1 pedicle.

■ Discussion

Fluoroscopically assisted pedicle screw placement is an accepted method of providing safe pedicle screw placement with efficacy established in a cadaveric study.[38] Guidelines to what radiologic anatomic boundaries should be used to assist safe screw passage have not been available.

Cortical markings have been used in other regions of complex anatomy to determine correlations between anatomy and radiology[27] and also in the thoracic spine when attempting to determine correlations of implants to pedicles.[8,35] It allows the determination of radiologic markings of significance, and it is obviously important when these markings are used for surgical procedures such as pedicle screw placement in the lumbosacral spine.

Pedicle dimensions in this study are consistent with those from previous studies described in the literature.[25,31,33,34,46] There has been some variation in published pedicle dimensions, yet it is accepted that the pedicles become wider in the lower lumbar spine. Olsewski et al[33] used measurements direct from cadaveric specimens and found results most compatible with those in this study. They found mean male pedicles had a height and width of 17.4 mm and 21.1 mm at L5, which is closely related to the values of 17.6 mm and 21.6 mm found in this study. Despite the agreement of the current results with those of predecessors in relation to height and width, the shape and dimensions of the L5 pedicle warrants further examination. The L5 pedicle has an elliptical shape similar to that of the pedicles above it, but the long axis of this ellipse becomes oblique to the vertical. This may explain why there is no observed increase in pedicle height at L5 when compared with more superior pedicles, along with the increase in pedicle width. Although the pedicle isthmus diameter at L5 is larger than at superior levels, it is considerably narrower than would be suspected from the pedicle width at L5 if the pedicle were of a round shape. Because the pedicle at L5 is elliptical, the long inferomedial boundary of the L5 pedicle is closely related to the L5 nerve root. This unique pedicle shape at L5, with an ellipse that has a longitudinal axis oblique to the vertical, may in part explain pedicle screw misplacement at this level.

It has been the authors' observation, although not directly reported in the literature, that articles and presentations frequently show pedicle screw misplacement as occurring at L5. This is surprising when the accepted dimensions of the L5 pedicle are the largest in the thoracic and lumbar spine. Possible reasons for this include a misunderstanding of the L5 pedicle shape, with its true elliptical shape being unrecognized (as described earlier), a much smaller than expected effective internal pedicle diameter,[31] the obliquity required to obtain access to the L5 pedicle axis requiring greater soft tissue retraction,[25] and the possibility of anatomic variation with resultant inaccuracy of anatomic entry point landmarks for the L5 pedicle.[36]

The description of the radiologic and anatomic correlations confirms that the cortical image of the pedicle obtained in coaxial fluoroscopy provides a safe guideline for positioning of pedicle screws. Placement of screws within the coaxial radiologic pedicle image allows a margin of safety, to prevent pedicle wall and associated neural damage. In this study there were no images that might misrepresent the true bony contour of the lumbosacral pedicles. The authors observed that often the image of the bony cortex of the pedicle (formed by the radiopaque wire) forms a halo around the coaxial pedicle image on the radiographs. The reason for this is unclear, but it may be that the inner pedicle wall appears as a greater density on the radiograph, separated from the radiographic marker by a relatively less dense outer pedicle wall. Similar observations of separation between cortical image and circumferential wire have been seen in other publications.[8,35]

At the sacrum, the image of the medial border of the S1 pedicle was found to be not the true pedicle but a trabecular line that continues out to the sacroiliac joint. Fortunately, this trabecular line consistently passes within the pedicle, and its image represents a safe guideline to the medial wall of the S1 pedicle. Because of the obliquity of the L5 and S1 pedicles the lateral pedicle border is occasionally not seen in the former and is never seen in the latter when viewed from the coaxial positions as described. However, other radiologic landmarks provide a safe guideline to the lateral border of both these

pedicles. At L5 the lateral image of the vertebral body is a guideline to the lateral border of the pedicle, and at S1 the lateral border of the pedicle is consistently lateral to the S1 foramen, which acts as a safe guideline. As other investigators have suggested, the transverse process base is not a reliable guideline to the vertical midpoint of the pedicle in the lumbar spine.[11]

■ Conclusions

In this study, the dimensions of the lumbar pedicles have been defined in the coaxial image and the oblique elliptical shape of the L5 pedicle has been shown, thus clarifying the anatomic dimensions of this pedicle. The results confirm that the L5 pedicle is less capacious than previously published measurements indicate. It also shows that the L5 pedicle wall has a long oblique inferomedial aspect, with the L5 root vulnerable to injury if any transgression of this wall occurs. The radiologic and cortical anatomy in the coaxial plane have been described, and it has been confirmed that the radiologic images of the pedicle anatomy represent safe guidelines for fluoroscopically assisted pedicle screw placement in this plane.

■ Key Points

- The radiologic pedicle image is a reliable guide to the lumbar pedicle anatomy in the coaxial plane.
- The S1 pedicle is visualized radiographically using both cortical images of the pedicle and other anatomic landmarks.
- The pedicle isthmic diameter at L5 is less than the pedicle width, and, therefore, the L5 pedicle is less capacious than previous measures have suggested.

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